A literature-implied negative answer for the BAP clause in Cuth–Doucha's totally disconnected question

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Source question

Cuth and Doucha ask at the end of *Lipschitz-free spaces over ultrametric spaces* [1]:

> Does the Lipschitz-free space over a totally disconnected compact metric space have the BAP? Is it a dual space?

This note records that the first clause has a negative answer by a later Cantor-metric theorem of Talimdjioski [2].

Supporting literature theorem

Let $K = 2^{\mathbb{N}}$ be the Cantor set and let $\mathcal{M}$ be the space of all metrics on K which induce the usual product topology. Talimdjioski defines

$$\mathcal{A}_f = \{d \in \mathcal{M} : \mathcal{F}(K, d) \text{ fails the approximation property}\}.$$

The theorem labeled `afmeagerdense` in [2] states that $\mathcal{A}_f$ is dense and meager in $\mathcal{M}$. In particular, $\mathcal{A}_f \neq \varnothing$.

Consequence for the source question

Corollary. *There exists a totally disconnected compact metric space M such that $\mathcal{F}(M)$ does not have the bounded approximation property.*

Proof. Choose $d \in \mathcal{A}_f$. Then $M = (K, d)$ is compact and totally disconnected, since d induces the usual Cantor topology. By the definition of $\mathcal{A}_f$, the Lipschitz-free space $\mathcal{F}(K, d)$ fails the approximation property. Since BAP implies AP for Banach spaces, $\mathcal{F}(K, d)$ cannot have BAP. □

Scope

This answers the BAP clause of the Cuth–Doucha/Godefroy question negatively. It does not settle the separate clause asking whether every free space over a totally disconnected compact metric space is a dual space. Indeed, [2, Corollary labeled `cor:dualmapresidual`] proves a residual set of compatible Cantor metrics for which $\mathcal{F}(K, d)$ is the dual of $\mathrm{lip}_0(K, d)$ and both spaces have MAP. Thus the later paper contains both positive generic information and negative existence information; the latter is the part used here.

References

[1] M. Cuth and M. Doucha, *Lipschitz-free spaces over ultrametric spaces*, arXiv:1411.2434.

[2] F. Talimdjioski, *Lipschitz-free spaces over Cantor sets and approximation properties*, arXiv:2305.07591.